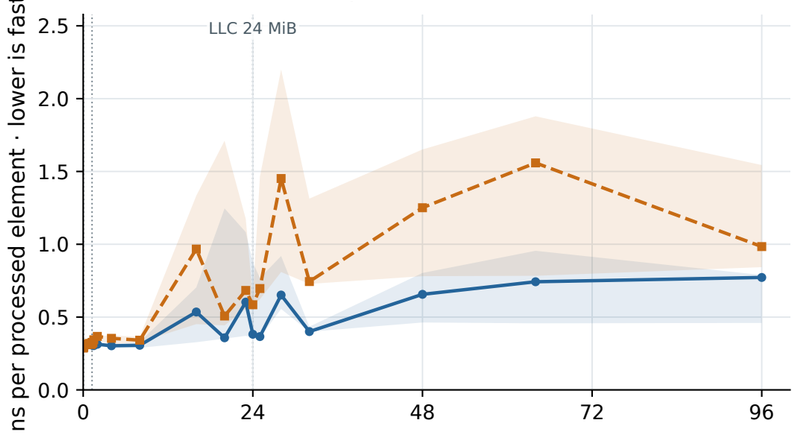
Float32 vs float64: time per processed element

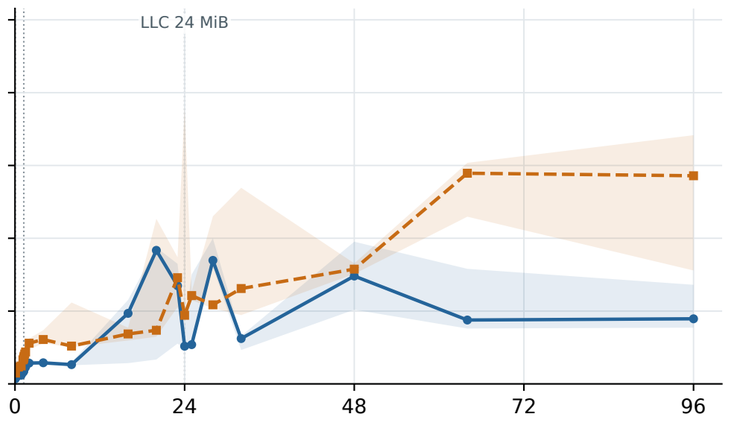
Shared axes within each panel · 34 measured working-set sizes per curve · same dense run 20261019T005422Z

float32 float64

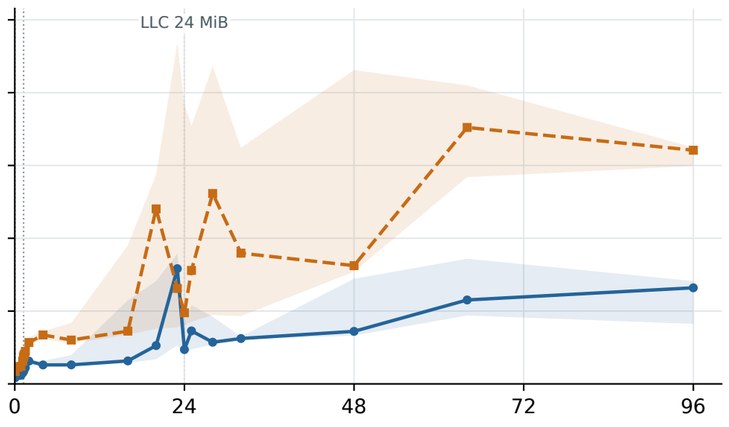
AXPY
Optimized scalar



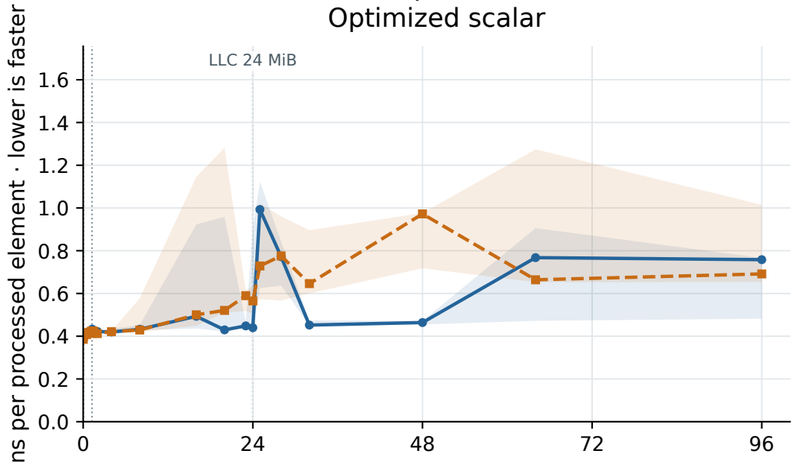
AXPY
Compiler auto-vectorized



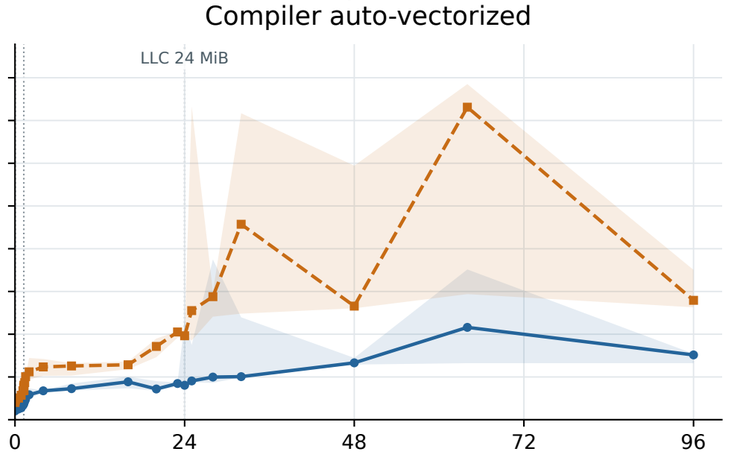
AXPY
Explicit AVX2 intrinsics



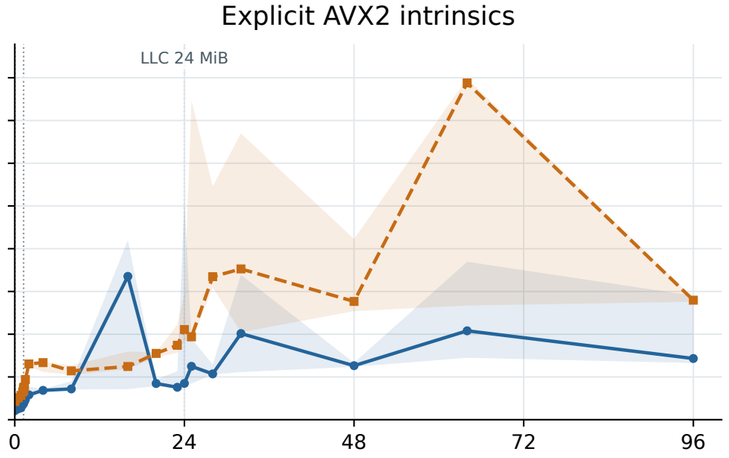
Dot product
Optimized scalar



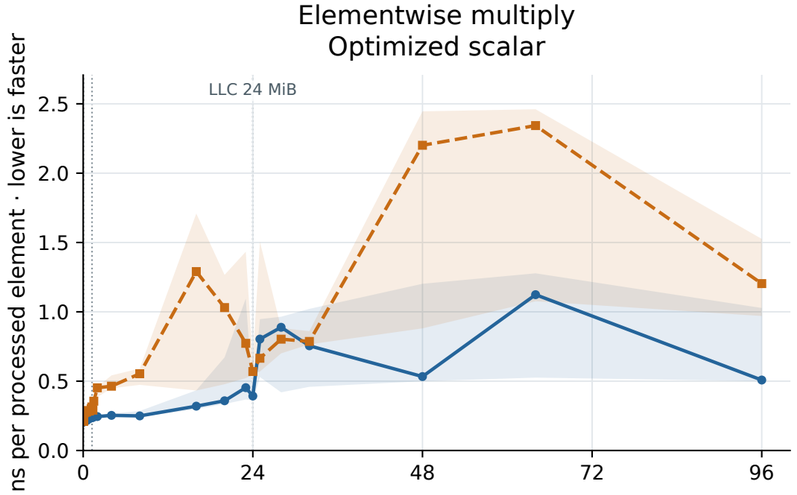
Dot product
Compiler auto-vectorized



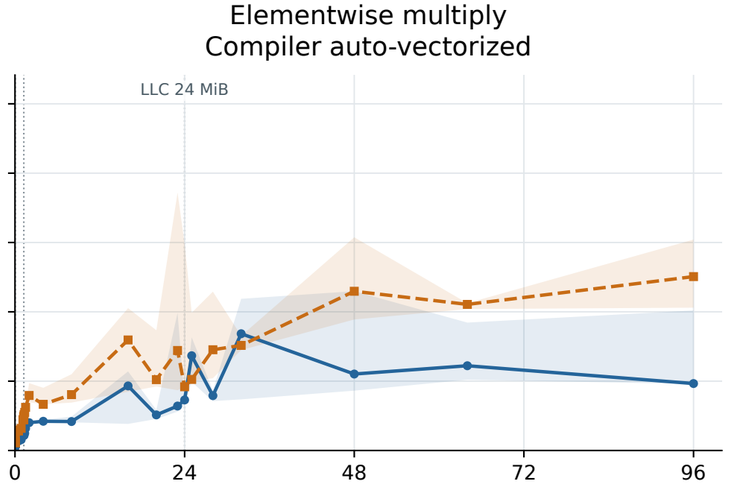
Dot product
Explicit AVX2 intrinsics



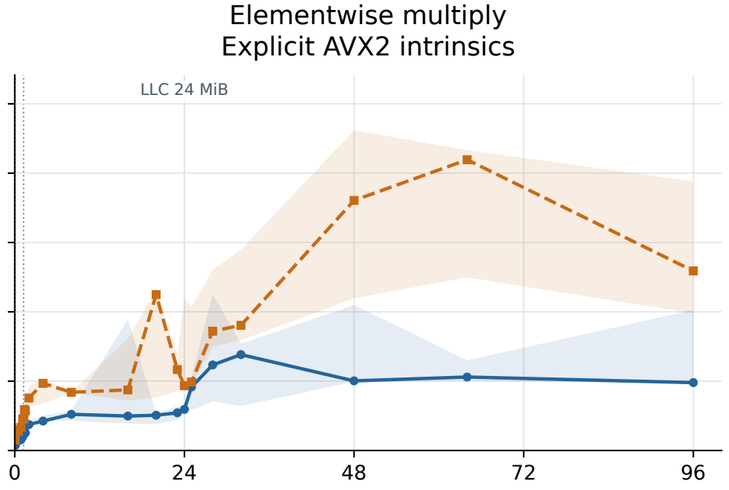
Elementwise multiply
Optimized scalar



Elementwise multiply
Compiler auto-vectorized



Elementwise multiply
Explicit AVX2 intrinsics



Total active working set across kernel arrays (MiB; linear scale)

Same nominal byte footprint, NOT equal N: float32 processes about twice as many elements. Actual byte footprints may differ by up to 96 bytes due to vector rounding.

Lines/markers: median of 3 process medians; shading: their observed range (not a CI). Every point uses 15 validated raw trials. No new measurements or interpolation.

CPU 0 pinned; same AVX2 / relaxed finite-input FP policy, stride 1 and aligned starts. DVFS/SMT not isolated. L1D/L2 marks cluster near zero; cache marks do not prove residency.